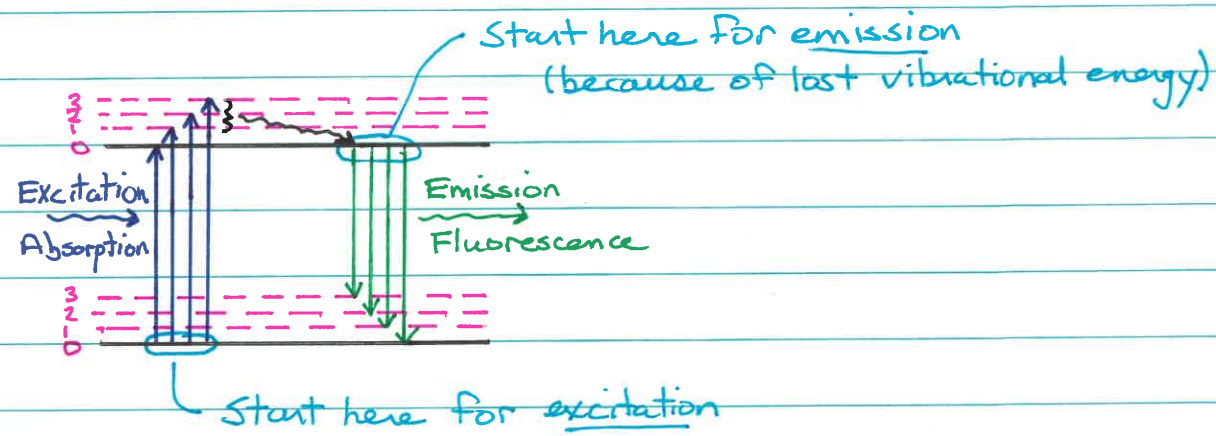
Lecture 14

2/20/26

Fluorescence Microscopy (continued)

Start w/ a brief review of the mechanisms underlying fluorescence, especially absorption & emission



There are two sources of relevant energy

- Electronic
- Vibrational

Excitation

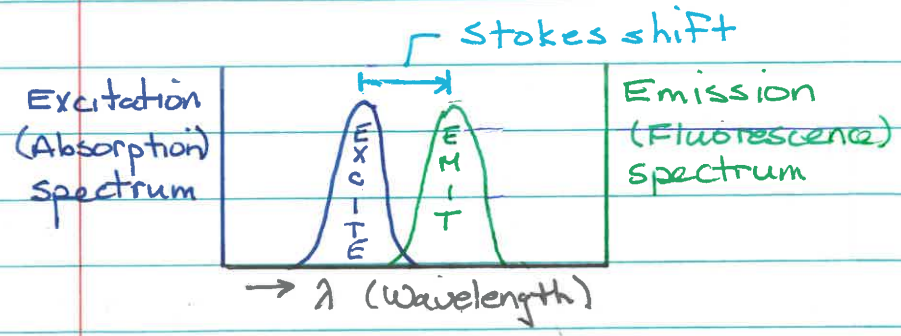
- Molecule always starts in the "lowest of the low" (i.e., it starts in the lowest vibrational energy level of the ground electronic state)
- Depending on the energy of the excitation photon, the molecule can get excited into a range of vibrational energy levels of the first excited electronic state

Emission

- Molecule always starts in the "lowest of the high" (i.e., it starts in the lowest vibrational energy level of the first excited electronic state). This is because the excited molecule very quickly ($\sim 10^{-12}$ sec) loses vibrational energy to the environment as heat via vibrational relaxation.
- The excited molecule can fall into a range of vibrational energy levels of the ground electronic state, releasing an emission photon of the corresponding energy.

Excitation & Emission Spectra

- Molecules absorb & emit photons over a range (or band) of energies, due to the effects of the vibrational energy levels.
- These "ranges" are reflected in the fluorophore's excitation (absorption) & emission spectra

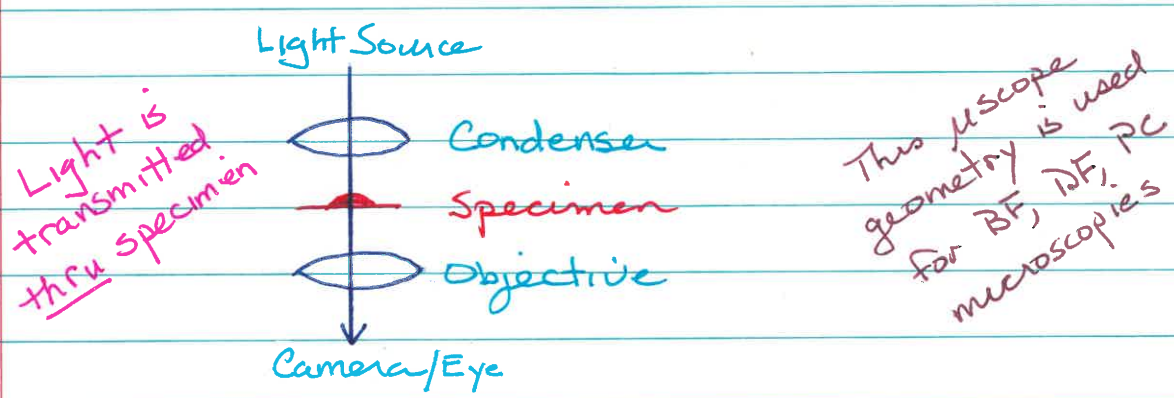


- Excite over range of λs
- Emit over range of λs
- Stoke's shift is λ difference between peak absorption and peak emission

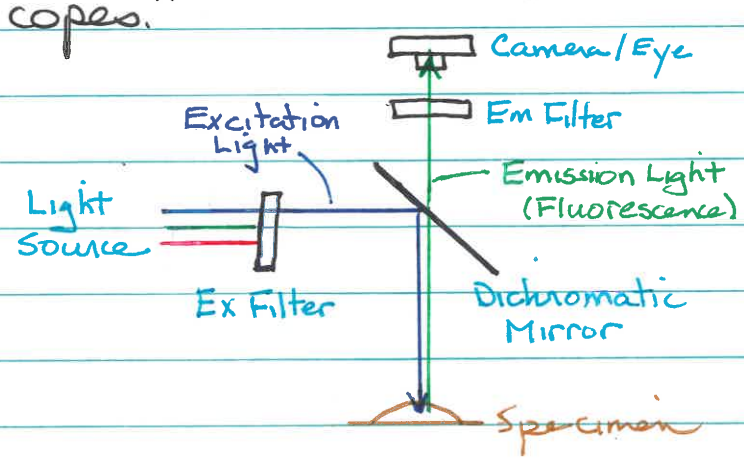
Typical fluorophores can be excited ~10^6 times before they photobleach.

Fluorescence Microscope

The techniques we have studied so far – brightfield (BF), darkfield (DF), & phase contrast (PC) – are transmitted light techniques.
They are called "transmitted light microscopy"



Fluorescence microscopes are different.
They are called "reflected light or epi-fluorescence microscopes."



Light Source

- Typically LEDs &/or lasers
- May emit particular wavelength(s) of light &/or white light consisting of a spectrum of wavelengths

Emission Filter

- Positioned between light source & dichromatic mirror
- Filters out light not needed to excite fluorescence
  In drawing, it filters out red & green light and transmits blue light
- Filters have very sophisticated construction & act by absorbing unintended colors
- Why hit fluorophore w/ colors it doesn't absorb?

Dichromatic Mirror

- Heart of the fluorescence microscope
- Reflects excitation light down toward sample (longer λ)
- Transmits emission (fluorescence) light toward detector (shorter)

Specimen

- Labeled w/ fluorophores
- In drawing, assume fluorophore = GFP { Absorbs BLUE, Emits GREEN

- In PowerPoint slide, fluorophore { Absorbs GREEN / Emits RED

Emission Filter

- Positioned between dichromatic mirror & detector
- If devices were perfect, would be able to send fluorescence straight to detector
- In practice, use an emission filter to remove any "stray" excitation light that might otherwise reach the detector & contaminate the signal
- Emission filter transmits emission (fluorescence) wavelengths & blocks other wavelengths
- In the drawing, it transmits green light & blocks blue light

Detector

- Camera, eye, etc. to form image

The goal in fluorescence imaging is to get rid of all excitation light so the image has a dark background

Key "ingredients"
- Stoke's shift
- Intelligent choice of filters
- Anti-parallel excitation & emission (so most excitation directed away from detector)

HINT: HW 6.9

Think, with these goals & attributes in mind, about what would happen if you switched the excitation & emission filters, removed the dichroic mirror, etc.

ACTIVITY 6 — Designing a Multi-Color Imaging Experiment

Sequential Imaging

Multi-color imaging is usually performed sequentially First image one fluorophore, then image next, ...

This is okay for fixed samples, but not great for living specimens

Simultaneous Imaging

Multi-color imaging can be performed simultaneously under certain conditions

- Excite - use two lasers at once
- Detect - want emission to be distinct enough to separate & detect simultaneously w/ two detectors

See completed Activity for solution

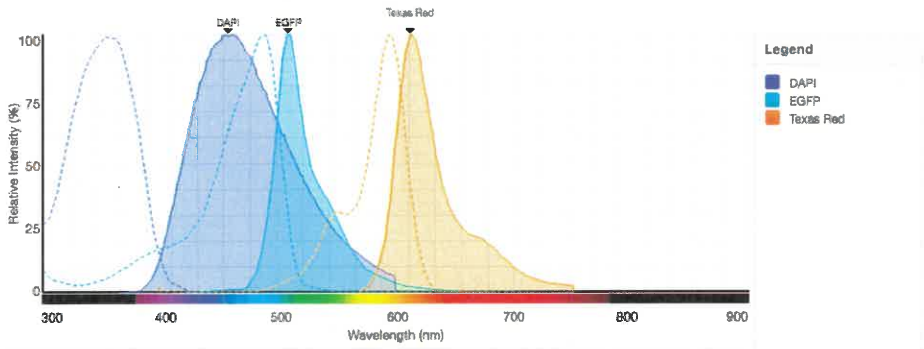
Name _____

Physics 390 In-Class Activity 6

Designing a Multi-color Imaging Experiment

Successful multi-color imaging requires generating a robust signal from each fluorophore and avoiding significant contamination of a signal from one fluorophore by signals from the other fluorophores. In the case of laser-based illumination, this involves choosing appropriate laser lines and emission filters and choosing between options that include simultaneous and sequential imaging.

Consider a sample labeled with the three fluorophores, DAPI, EGFP, and Texas Red. The excitation (dashed lines/unfilled areas) and emission (solid lines/filled areas) spectra for these three fluorophores are shown below. To generate robust signals, you want to choose a laser line that overlaps well with each fluorophore's excitation spectrum. Similarly, you want to choose an emission filter that will pass a significant fraction of each fluorophore's emission with (hopefully) minimal signal contamination from other fluorophores.



Unnormalized excitation and emission spectra

Key
Excitation: - - - Emission: ——

Your laser line options are:

405 nm, 458 nm, 488 nm, 514 nm, 561 nm, and 633 nm.

Your emission filters options are:

350/50 nm, 450/50 nm, 490/50 nm, 525/50 nm, 590/50 nm, and 615/50 nm.

Note: The notation 350/50 nm means that the filter passes EM radiation with wavelength ranging from 325 to 375 nm (i.e., 25 nm on either side of 350 nm).

a) Choose the best excitation line for each of the three fluorophores and enter your choice into the table. Note: DAPI is the hardest to excite with these laser lines, but one choice will work with low efficiency.

Fluorophore	Excitation Line	Emission Bandpass	
DAPI	405 nm*	450/50 nm	425-475 nm
EGFP	488 nm†	525/50 nm	500-550 nm
Texas Red	561 nm	615/50 nm	590-640 nm

b) Choose the best emission filter set for each fluorophore and enter your results into the table.

* Poor choice but best option
† Best matched to absorption peak

c) In general, fluorophores in multi-color samples are imaged sequentially. Sequential imaging involves directing only one laser line onto the sample at a time, collecting the resulting fluorophore emission, and then repeating this process for the other fluorophores. For living samples, it can be preferable to image fluorophores "simultaneously," which involves using multi-laser excitation and then collecting the emission simultaneously using a different detector for each fluorophore. Avoiding significant contamination of a signal from a fluorophore by signals from other fluorophores is much more difficult to achieve with simultaneous imaging, which is one reason this approach is less commonly used.

For each fluorophore combination below, determine if simultaneous and/or sequential imaging is viable with your filters and place a "check mark" in the table when the method is an option.

Fluorophore	Simultaneous	Sequential
DAPI/EGFP	No*	✓
DAPI/Texas Red	✓	✓
EGFP/Texas Red	✓	✓
DAPI/EGFP/Texas Red	No*	✓

* Too much spectral overlap